

File Edit Control Applications ?



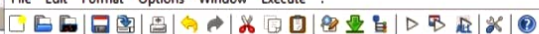
Scilab 2025.1.0 Console

```
"Spherical to Cartesian Coordinates (x, y, z):"  
2.3492316 4.0689884 1.7101007  
"Spherical to Cylindrical Coordinates (p, phi, z):"  
4.6984631 60. 1.7101007
```

-->

exp case 2.sci (C:\Users\pachi\Documents\exp case 2.sci) - SciNotes

File Edit Format Options Window Execute ?



exp case 2.sci (C:\Users\pachi\Documents\exp case 2.sci) - SciNotes

exp case 3.sci X exp case 2.sci X

```
1 clc;  
2 clear;  
3  
4  
5 r = 5;  
6 phi = 60;  
7 theta = 70;  
8  
9  
10 x = r * sind(theta) * cosd(phi);  
11 y = r * sind(theta) * sind(phi);  
12 z = r * cosd(theta);  
13  
14 disp("Spherical to Cartesian Coordinates (x, y, z):");  
15 disp([x y z]);  
16  
17  
18 p = r * sind(theta);  
19 phi_cyl = phi;  
20 z_cyl = r * cosd(theta);  
21  
22 disp("Spherical to Cylindrical Coordinates (p, phi, z):");  
23 disp([p phi_cyl z_cyl]);  
24
```

Ques iii)

~~$r = 10$~~ ;

$$r = 5$$

$$\theta = 60^\circ$$

$$\phi = 70^\circ$$

$S \rightarrow C$

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$x = 5 \times 0.9397 \times 0.5 = 2.349$$

$$y = 5 \times 0.9397 \times 0.866 = 4.0681$$

$$z = 5 \times 0.3420 = 1.7101$$

S-7C

$$p = r \sin \theta$$

$$\phi = 60^\circ$$

$$z = r \cos \theta$$

$$p = 5 \times 0.9397 = 4.6985$$

$$z = 5 \times 0.3420 = 1.7101$$

$$\phi = 60^\circ$$